Adarsh Kumar Jha

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Github | Linkedin

Education

Indraprastha Institute of Information Technology, Delhi **CGPA: 8.09/10**
B.Tech (Computer Science and Artificial Intelligence)
Nov 2022 - Jun 2026

Kendriya Vidyalaya Pitampura, Delhi **Percentage: 92%**
CBSE, Standard 12, PCM + CS(Elective)
Apr 2021 - Jun 2022

Kendriya Vidyalaya Pitampura, Delhi **Percentage: 89%**
CBSE- Standard 10
Apr 2019 - Mar 2020

Skills

Programming Languages: Python, Java, C/C++, Javascript, MATLAB, Swi-Prolog

Tools and Technologies: Langchain, Huggingface, Docker, ROS, Git/GitHub, AWS, Cursor, Linux, Figma, MySQL Workbench, Kaggle, Wireshark, Matplotlib, Yolo, Google Colab, Ubuntu (WSL), NS-3, AI Agents, Retrieval Augmented Generation (RAG)

Frameworks: PyTorch, TensorFlow, Pandas, Scikit-learn, JavaFX, React, REST API, FastAPI

Technical Electives: Data Structures & Algorithms, Machine Learning, Natural Language Processing, Computer Vision, Object Oriented Programming (Java), Computer Networks, Operating System (C), Cybersecurity, DBMS (MySQL), Convex Optimization, Signals and System, Linear Algebra, Probability & Statistics, Algorithm Design and Analysis, Artificial Intelligence, Data Science

Work Experience

ML Engineer – Research to Commercialization *May 2025 - Jul 2025*
Advisor: Dr. Anuj Grover **r2c.ai.iiitd.in**
`PyTorch, Transformers, FastAPI, Docker, React.js, Node.js, LoRA`

- Led the end-to-end creation of a Transformer-based Q&A model, mitigating hallucinations, resulting in **95% factual accuracy** while using 8-bit quantization and LORA to cut model size by **75%** and **double inference speed**.
- Architected and deployed a scalable full-stack application using FastAPI, Docker, and React.js to handle **100+ concurrent requests**, integrating a layout-aware OCR pipeline that improved text extraction accuracy by **15%**.

Research Intern – Ontology Path Extraction with Oracle *Jan 2024 - May 2024*
Advisor: Prof. Raghava Mutharaju (IIIT-Delhi) and Suman Roy (Oracle)
`Transformers, LLaMA, Mistral, GPT-4, Huggingface, R-BERT, Knowledge Graphs`

- Devised a novel R-BERT pipeline, **surpassing LLaMA-3, Mistral, and GPT-3** performance on biomedical datasets by **6% in F1-score**, using span-based embeddings and negative sampling for enhanced relation extraction.
- Mined relationships between **25,000+ biomedical entities** using a novel R-BERT pipeline, resulting in a **20% reduction in false positives** during complex query interpretation for clinical trial matching.

Projects

Deception Detection in Diplomacy Dialogues *Jan 2025 – Apr 2025*
`Python, TensorFlow, GCNs, NLP, Transformers, Bi-LSTM` **Team Size: 2** | **GitHub**

- Architected and progressively refined a series of models for deception detection, beginning with a Bi-LSTM baseline and advancing to a final hybrid architecture that fused a Transformer text encoder with a Graph Convolutional Network (GCN) to model the game's social dynamics.
- Preprocessed a dataset of over 17,000 messages, using down-sampling to address class imbalance, and attaining **85.8% accuracy** with a **0.60 Macro F1-score** on the held-out test set.

Watch-Tower S3R: Real-Time Video Anomaly Detection *Jan 2025 – Apr 2025*
`Python, PyTorch, Flask, OpenCV, I3D` **Team Size: 2** | **GitHub**

- Incorporated LoRA and quantization to a Self-Supervised Sparse Representation (S3R) model paired with an I3D feature extractor, reaching a frame-level **AUC of 83.5%** on the UCF-Crime dataset.
- Cut model inference time by over **28%** (from 1.20s to 0.86s) by optimizing the attention mechanism bottleneck, with the system's capabilities demonstrated through a full-stack application with a live monitoring dashboard.

Mini Operating System *Jul 2023 – Aug 2023*
`C, Linux` **Team Size: 2** | **GitHub**

- Designed and built a custom Linux shell in C, implementing robust command execution, persistent command history, auto-completion, and advanced signal handling.
- Mastered core OS concepts by implementing a priority-based CPU scheduler to optimize process management, alongside systems for process control (fork, exec), file I/O, and Inter-Process Communication (IPC).

E-Commerce Website *Jan 2024 – May 2024*
`Python, JavaScript, HTML, CSS, SQL` **Team Size: 3** | **GitHub**

- Built a responsive, full-stack e-commerce platform featuring robust user authentication, cart management, order history, and an administrator dashboard, which led to an **88% task success rate** in user testing.
- Applied a normalized SQL database schema and implemented rigorous transaction controls that safeguarded data integrity by eliminating **100% of concurrency conflicts** during high-traffic simulations.

Stick Hero Game *Oct 2023 – Dec 2023*
`Java, JavaFX, Scene Builder, Design Patterns, JUnit` **Team Size: 2** | **GitHub**

- Engineered a Stick Hero clone using Java, JavaFX, and Scene Builder with a responsive UI and smooth animation. Improved player retention by 20% through optimized mechanics and user experience.
- Used Singleton and Factory design patterns, leveraging multithreading for **10+ concurrent animations** and ensuring reliability with JUnit tests that achieved **90% path coverage**.

Positions of Responsibility

- **Events Head**, Esummit IIIT Delhi: Directed the end-to-end planning and execution of all workshops and flagship competitions for the annual entrepreneurship fest. *Feb 2025 - Mar 2025*
- **Operations Team Head**, Odyssey IIIT-D: Led a 30-member team managing logistics and vendor coordination for the annual cultural fest with a 30,000+ footfall. *Jan 2025 – Mar 2025*

Awards and Achievements

- Solved 300+ coding problems across platforms: LeetCode, Codeforces, and GeeksforGeeks.
- **JEE Mains 2022:** Secured **AIR 4778 (99.5 percentile)** among 872,970 students and cleared JEE Advanced.
- **Indian National Physics Olympiad (INPhO):** Ranked in the **top 5% nationally** (High Distinction).
- **Chess.com Legend:** Achieved Legend Title after 200+ hours of strategic gameplay.

Certifications

- **Ethical Hacking Essentials** - Coursera `Web Attacks, IoT, OT, Cloud Computing, Pentesting`
- **Machine Learning Specialization** - Coursera `Regression, Neural Networks, Clustering`
- **Data Structures and Algorithms** - Udemy `Trees, Graphs, Hashmaps, Algorithms`

Interests and Hobbies

Engage in competitive programming and tournament chess; also enjoy table football.

Declaration: The above information is correct to the best of my knowledge.
Adarsh Kumar Jha
Date: July 16, 2025